Section 4: Fuel Usage, Outlier Removal and Fuel Efficiency

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Abstract

This section covers outlier removal for the five most common currencies in the fuel logbook dataset (Section 4.1) and six of the fuel efficiency questions that follow from it (Section 4.2, questions 1, 2, 3, 4, 8 and 9). All figures and statistics below are computed on the full, cleaned dataset of 1,173,702 fuelling records produced by Sections 1–3.

1 Outlier Removal

1.1 Top 5 currencies by transaction count

The five most common currencies by number of transactions are `$` (USD), `£` (GBP), `€` (EUR), `CA$` (CAD) and `R` (ZAR), as shown in Table 1. As anticipated in the assignment brief, the Rand is the fifth most common currency.

Table 1: Top 5 currencies by transaction count.

Currency	Transactions
`$` (USD)	741,655
`£` (GBP)	87,554
`€` (EUR)	59,254
`CA$` (CAD)	46,814
`R` (ZAR)	36,388

1.2 Per-currency outlier removal

The brief's own example, a South African user tagged with `$` reporting several hundred dollars for only a few tens of litres, is really a *cost-per-litre sanity check*: a fill-up's implied price (`total_spent / litres`) has to sit within the historically plausible retail fuel price band for that currency. Values wildly outside that band point either to a data-entry error (stray digits, a decimal point in the wrong place) or a currency tag that does not match the price actually paid.

For each of the top 5 currencies, a row is kept only if:

1. `total_spent` and `litres` are present and `total_spent` > 0;
2. `litres` falls in $[1, 300]$: below 1 L is not a real fill-up, and 300 L comfortably covers even large trucks and vans, so this only removes clear entry errors (e.g. an odometer reading captured in the gallons field);

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3. the derived cost per litre falls inside a currency-specific plausible band, chosen generously to span 2005–2022 retail prices, including the 2021/22 price spike, while still excluding gross misclassifications (Table 2).

These bounds were checked against the empirical percentiles of each currency: in every case the 95th/99th percentile sits comfortably inside the band, while the 99.9th percentile jumps by one to two orders of magnitude: a clear break between genuine price variation and garbage data.

Table 2: Plausible cost-per-litre bounds used for outlier removal (local currency per litre).

Currency	Band
USD ($)	0.20 – 3.00
GBP (£)	0.50 – 3.00
EUR (€)	0.50 – 3.00
CAD (CA$)	0.50 – 2.50
ZAR (R)	6.00 – 24.00

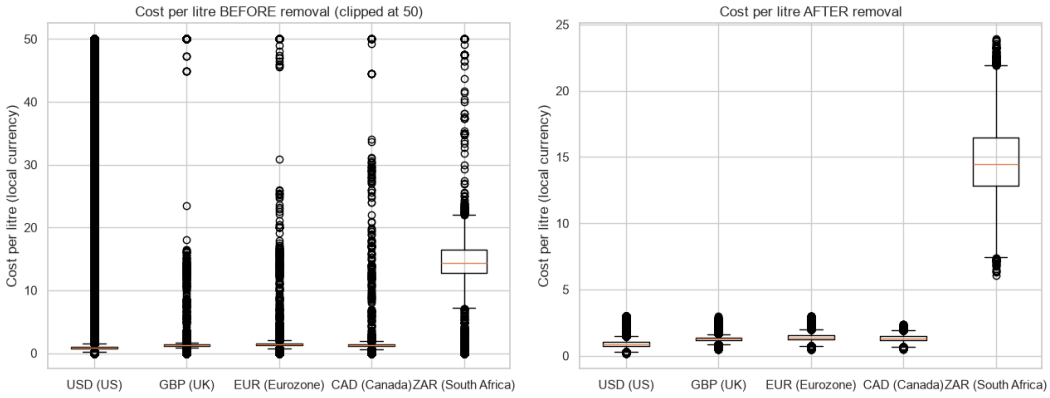


Figure 1: Cost per litre per currency before (left) and after (right) outlier removal. The pre-removal panel is clipped at 50 for legibility; the uncapped tail reaches several orders of magnitude higher.

1.3 Rows removed

Table 3 shows the number of rows removed per currency. In total, 55,759 of 971,665 rows (5.74%) across the top 5 currencies were removed as outliers, leaving 915,906 clean rows for the fuel efficiency analysis in Section 2. Removal rates range from 3.03% (CAD) to 6.32% (USD), low enough that the bounds are not simply discarding genuine variation, and consistent across currencies, which suggests the bands in Table 2 are reasonably well calibrated rather than arbitrarily strict for any one currency.

Table 3: Rows removed per currency.

Currency	Before	Removed	% removed
USD (US)	741,655	46,903	6.32
GBP (UK)	87,554	4,031	4.60
EUR (Eurozone)	59,254	2,201	3.71
CAD (Canada)	46,814	1,420	3.03
ZAR (South Africa)	36,388	1,204	3.31
Total	**971,665**	**55,759**	**5.74**

2 Fuel Efficiency

All analysis in this section uses the 915,906-row dataset remaining after the outlier removal described in Section 1.

2.1 Q1: Cost per litre per country, January 2022, converted to ZAR

Average January-2022 exchange rates to ZAR were computed from Nedbank's published daily rates for the 21 trading days in January 2022 (USD/ZAR, GBP/ZAR and EUR/ZAR); CAD/ZAR was derived as USD/ZAR ÷ USD/CAD, using a USD/CAD rate of 1.263 sourced from valutafx.com and ofx.com for the same month (Table 4).

Table 4: Exchange rates to ZAR used for Q1 (average, January 2022).

Pair	Rate (1 unit = X ZAR)	Source
USD/ZAR	15.53	Nedbank daily rates, 3–31 Jan 2022 (21 trading days)
GBP/ZAR	21.06	Same source
EUR/ZAR	17.59	Same source
CAD/ZAR	12.30	Derived: USD/ZAR ÷ USD/CAD (USD/CAD $\approx$ 1.263)
ZAR/ZAR	1.00	Identity

Table 5: Average cost per litre, January 2022, converted to ZAR.

Country	n	Local mean	ZAR per litre
United Kingdom (GBP)	1,113	1.481	31.20
Eurozone (EUR)	1,243	1.586	27.91
South Africa (ZAR)	990	18.951	18.95
Canada (CAD)	1,782	1.473	18.11
United States (USD)	36,620	0.951	14.78

The UK and Eurozone show the highest ZAR-equivalent litre price, South Africa and Canada sit in the middle, and the US is the cheapest. This tracks fuel taxation rather than crude oil cost: US federal and state fuel tax is comparatively low, while UK and EU fuel duty and VAT are among the highest in the world. South Africa's price sits above the US/Canada level mainly because of the weaker Rand and the fuel levy and Road Accident Fund levy stacked on the base price, not because crude is more expensive there.

2.2 Q2: Detecting missed (unlogged) fill-ups

Rule. For a given user, fill-ups are sorted by date and the odometer gap between consecutive logged entries is computed. If every fill-up were logged, that gap should be reasonably close to the user's own *typical* refuelling distance. A gap much larger than that user's own median gap means at least one fill-up in between almost certainly went unlogged. Comparing each gap to the same user's own median, rather than a single global threshold, automatically accounts for the fact that different people drive very different annual distances. Users need at least 3 valid gaps for their median to be meaningful. A gap $> 2\times$ that user's median gap is flagged as a likely missed fill-up.

Plotting the ratio of each gap to that user's own median gap gives a distribution heavily concentrated just above 1.0 (most fill-ups happen close to a driver's usual rhythm), with a long right tail past the 2.0x threshold rather than a second, separate peak, so the cutoff is a judgement call rather than a natural break in the data. Of 681,449 evaluated fill-up gaps, 120,248 (17.65%) exceed twice that user's median gap and are flagged as containing at least one missed fill-up. As a sensitivity check, 11.92% of gaps exceed a 2.5x threshold and 7.99% exceed 3.0x. Roughly 1 in 6 consecutive fill-up pairs shows a gap more than double that driver's usual refuelling distance, i.e. a fill-up almost certainly happened in between without being logged. This is a lower bound: a user who misses two fill-ups in a row, or who fills up partially every time, will not necessarily clear a 2x threshold.

2.3 Q3: Average distance per tank, per country

Table 6: Average distance per tank (km) by country.

Country	Mean (km)	Median (km)
Eurozone (EUR)	593.3	574.3
United Kingdom (GBP)	577.2	553.2
South Africa (ZAR)	541.2	523.0
United States (USD)	458.5	444.3
Canada (CAD)	441.6	426.9

The Eurozone has the largest average distance per tank (593 km). Eurozone and UK drivers lead, ahead of South Africa, with the US and Canada lowest. This lines up with vehicle mix rather than tank size: Europe and the UK have a much higher share of diesel vehicles, which are materially more fuel-efficient per litre than the petrol-dominant US/Canadian fleets, so a similarly-sized tank simply carries those vehicles further.

2.4 Q4: Do newer vehicles drive further between fill-ups?

Yes: distance per tank falls steadily and close to monotonically with vehicle age (correlation $= -0.176$), from approximately 510 km for cars 0–3 years old down to approximately 376 km for cars over 25 years old, across every age bracket tested in between. This is consistent with newer vehicles generally having better fuel efficiency and often larger tanks, and with older vehicles losing some efficiency (wear, outdated engine technology) over time.

2.5 Q8: Correlation between fuel efficiency and other features

Fuel efficiency (`litres_per_100km`) is itself computed as `litres / killometre * 100`, so `litres`, `total_spent` and `cost_per_gallon` are excluded from this analysis: correlating a value against a term used to derive it is not a genuine relationship, it is circular. The real candidates are distance (`killometre`), vehicle age/year, and vehicle model/make.

`car_model` (1,945 levels) and `car_make` (29 levels) are not numeric, so each is converted to a numeric signal via 5-fold mean-target encoding: a row's encoded value is the average efficiency of *other* rows sharing its model or make, never its own. This keeps the categorical relationship usable without a row leaking into its own encoded value.

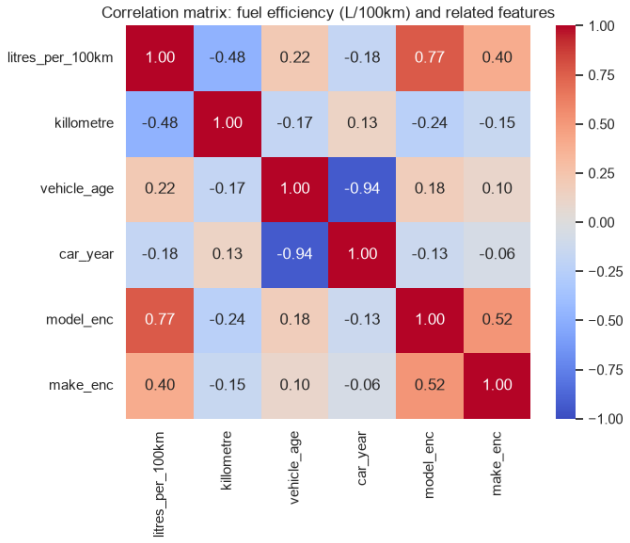


Figure 2: Correlation matrix: fuel efficiency (L/100km) and related features.

This confirms the expected pattern: model (0.77) is the strongest correlate by far, distance travelled is next (-0.48; more distance per tank implies a more efficient or larger-tanked car), make is moderate (0.40), and vehicle age is weaker (0.22). `car_year` and `vehicle_age` are almost perfectly anti-correlated (-0.94), since `vehicle_age = fueling_year - car_year` and `fueling_year` varies far less than `car_year`: the two columns carry almost the same information twice.

2.6 Q9: Random Forest feature importance

A Random Forest was trained on a 200,000-row sample using the same five numeric/encoded features as Q8, excluding `litres_per_100km`'s own algebraic inputs for the same reason as above (train $R^2 = 0.783$).

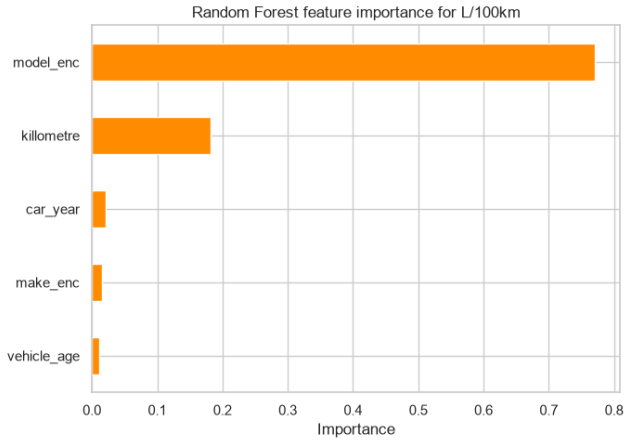


Figure 3: Random Forest feature importance for L/100km.

The Random Forest agrees with the correlation ranking on the top feature: model dominates (importance ≈ 0.77), though it diverges further down. Distance keeps a clear second place (importance ≈ 0.18), much as its correlation magnitude suggested. `car_year`, `make_enc` and `vehicle_age` all drop to under 0.02 each, well below what their correlation coefficients (-0.18, 0.40, 0.22) would suggest on their own. Two things explain the gap. First, `car_year` and `vehicle_age` are almost perfectly collinear, so the forest can use either one and importance gets split between them rather than credited fully to one. Second, `make_enc` is itself largely explained by `model_enc` (0.52 correlation between the two, since make is a coarser version of model), so once the forest has split on model there is little independent signal left in make. Correlation looks at each feature in isolation; the Random Forest looks at marginal contribution once the other features are already in the tree, so redundant/collinear features naturally look far less important there even if their raw correlation is moderate.